

SINGmaster: An Android app that uses TarsosDSP

From version 2.0 onwards, the framework is Android-compactable. SINGmaster is the first Android application developed using TarsosDSP. This smartphone application helps a user to learn how to sing. There are practical exercises based on the important building blocks of melodies. The application makes use of pitch-tracking capabilities of TarsosDSP to give visual feedback, which allows the user to learn from mistakes.

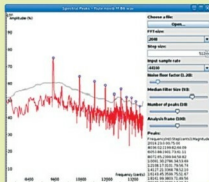


Fig. 3: Visualisation of spectral peaks of a flute

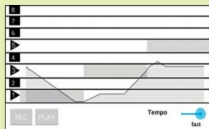


Fig. 4: Screenshot of SINGmaster application

sis analysis as well as synthesis, and is based on C language.

Create Signal Library (Sizzle) and SndObj are more C++ based projects that allow users to have both audio analysis and synthesis in real-time.

Beads, JASS, JSyn and Minim are Java based projects that allow users to have real-time audio synthesis.

TarsosDSP is introduced as a single framework that could cater real-time music analysis and synthesis needs in Java—something that no other competing tools provide—as per the research conducted by the developers of the tool.

Although algorithms used in software like jAudio and YAAFE are more efficient due to the reuse of calculations for feature extraction, these are less readable compared to TarsosDSP.

Highly-optimised libraries such

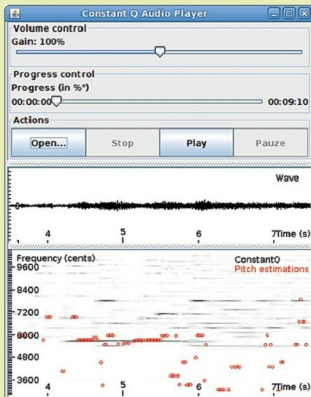


Fig. 5: Constant-Q transformation with an overlay of pitch estimations and corresponding waveform

as SoundTouch are again less beneficial for beginners on account of low readability.

Main features implemented

The tool was originally devised as a library for pitch estimation. Therefore you will find several pitch-estimating algorithms in this framework including YIN, McLeod Pitch Method (MPM), Average Magnitude Difference Function (AMDF) and an estimator based on dynamic wavelets.

There is a time-stretch algorithm using which the user can alter the speed of audio without changing the pitch.

Alternatively, we also have a pitch-shifting algorithm that allows the user to change the pitch without

changing the speed. A resampling algorithm with related software package is also included.

TarsosDSP consists of two onset detectors and several infinite impulse response (IIR) filters including low-pass, high-pass and band-pass filters. Audio effects like delay and flanger effect are also provided. In a flanger, two copies of the same signal are mixed, with one of these slightly delayed for not more than 20 milliseconds to produce a swirling effect.

Another important functionality of the tool is audio synthesis. We can find sine-wave and noise generator in the tool. A low-frequency oscillator is also incorporated for amplitude control. Spectrum-estimating algorithms based on fast Fourier transforms and other

techniques are also available.

Lossy audio compression using Haar Wavelet Transform

Haar Wavelet Transform (HWT) is widely used in various image- and signal-processing applications. It could be effectively used in signal compression or for edge detection in images. Let us take a look at one such use case of HWT algorithm implemented in TarsosDSP.

HWT used for a simple Lossy audio compression consists of the following steps:

First, it compresses the audio by dividing it into blocks of 32 samples. These are then transformed using HWT, and samples with the least difference between these are